

## Highlights

### **Tree Ensembles and Tabular Transformers for Financial Fraud Detection: Imbalance Sensitivity, Controlled Transfer and Interpretation Boundaries**

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- Boosted trees lead transfer point estimates on five filtered BAF cohorts
- FT-Transformer loses about 35% average precision with primary SMOTE
- A categorical-aware control does not recover the transformer baseline
- SHAP set agreement does not establish rank stability or causality
- Diffuse final-layer attention is a separate descriptive diagnostic

# Tree Ensembles and Tabular Transformers for Financial Fraud Detection: Imbalance Sensitivity, Controlled Transfer and Interpretation Boundaries

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## ABSTRACT

Close in-domain scores do not establish that fraud models respond similarly to imbalance interventions or distribution shift. We compare classical models and a custom FT-Transformer through imbalance sensitivity, frozen Base-to-Variant transfer, compatible SHAP analysis and a separate attention diagnostic. The ULB experiment separates distinct predictor profiles before splitting. BAF transfer excludes exact matches to Base development profiles from each Variant test cohort, retaining 177,526–178,828 observations under a common 30-predictor projection. Preprocessing and thresholds are selected within development data. On BAF Base, CatBoost, FT-Transformer and LightGBM baseline average precision is 0.1796, 0.1769 and 0.1768, respectively. Primary SMOTE reduces FT-Transformer average precision by approximately 35%, compared with 14% for CatBoost. A predefined categorical-aware FT-Transformer control reaches 0.1157, not the 0.1769 baseline; the comparison does not isolate an architectural cause. LightGBM leads transferred average precision on Variant I and CatBoost on Variants II–V. Every supervised model has lower transferred  $F_2$  than on Base under its frozen threshold. The boosted models share a top-10 SHAP feature set, and the fixed LightGBM model retains that set across the evaluated BAF cohorts, without identical rankings or guaranteed explanation validity. FT-Transformer final-layer, head-averaged attention is diffuse (normalised entropy 0.985), a descriptive result rather than evidence of causality or an information ceiling. Together, these analyses support evidence-bounded model selection while keeping representation, operating-point and interpretation limitations explicit.

## 1. Introduction

Financial fraud detection combines rare positive cases with an operational trade-off between missed fraud and false alerts (Ryman-Tubb, Krause and Garn, 2018; Hilal, Gadsden and Yawney, 2022). Choosing a model therefore requires more than comparing one in-domain score. A useful evaluation also asks how imbalance interventions affect the implemented pipeline, how a frozen model behaves on changed populations, and what its inspected explanations support.

Gradient-boosted trees remain strong tabular baselines (Ke, Meng, Finley, Wang, Chen, Ma, Ye and Liu, 2017; Prokhorenkova, Gusev, Vorobev, Dorogush and Gulin, 2018; Shwartz-Ziv and Armon, 2022), while attention-based models such as FT-Transformer offer a different representation of feature interactions (Gorishniy, Rubachev, Khrulkov and Babenko, 2021). Benchmark and survey evidence motivates dataset-dependent comparison rather than a universal architectural ordering (Grinsztajn, Oyallon and Varoquaux, 2022; Borisov, Leemann, Seßler, Haug, Pawelczyk and Kasneci, 2024). Our unpublished companion manuscript (Caixeiro, Abbasi and Martins, 2026) examines baseline performance and validation-derived decision thresholds. The present article uses the same pinned evidence but focuses on sensitivity, controlled transfer and the boundaries of interpretation.

We make four contributions:

**C1. Representation-sensitive imbalance comparison.** FT-Transformer and CatBoost are compared across seven predefined strategies on ULB and BAF, with categorical-aware BAF controls reported separately.

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- C2. **Controlled transfer on audited cohorts.** Base-fitted models and thresholds are frozen before evaluation on five BAF Variant cohorts excluding exact common-predictor matches to Base DEV.
- C3. **Compatible SHAP set comparisons.** Linear and boosted-tree explanations are compared at the original-feature level; the fixed Base-trained LightGBM is also inspected on retained Variants.
- C4. **Separate attention diagnostics.** Final-layer, head-averaged FT-Transformer attention is described without treating it as faithful attribution or a test of a dataset-imposed ceiling.

The contribution is an integrated, traceable evaluation, not a causal account of architectural superiority. It distinguishes ranking performance from threshold-dependent performance, representation effects from architecture, and global feature-set agreement from ranking or causal stability. Section 2 reviews the methods; Section 3 defines the protocol; Section 4 presents the evidence; Sections 5 and 6 discuss its implications.

## 2. Related Work

### 2.1. Tabular modelling and imbalance interventions

LightGBM and CatBoost are distinct gradient-boosting implementations (Ke et al., 2017; Prokhorenkova et al., 2018); Random Forest combines randomised trees (Breiman, 2001). FT-Transformer tokenises numerical and categorical inputs before transformer processing (Gorishniy et al., 2021). The comparative literature identifies strong tree baselines and dependence on datasets and preprocessing choices (Shwartz-Ziv and Armon, 2022; Grinsztajn et al., 2022; Borisov et al., 2024). Those findings motivate comparison but do not predict a universal winner under fraud-specific interventions.

SMOTE synthesises minority observations by interpolation (Chawla, Bowyer, Hall and Kegelmeyer, 2002); cleaning combinations add Tomek-link or edited-nearest-neighbour stages (Batista, Prati and Monard, 2004). The representation on which interpolation operates matters. Interpolating one-hot columns or ordinal category codes differs from preserving nominal category membership. We report the executed preprocessing and sampling paths rather than attributing all differences to architecture.

### 2.2. Controlled distribution transfer

The BAF suite provides a synthetic framework for examining imbalance, bias and distribution changes (Jesus, Pombal, Alves, Cruz, Saleiro, Ribeiro, Gama and Bizarro, 2022b,a). Its Variants share a generating process with Base, making them useful controlled stress tests but not independent institutional deployment studies. Freezing Base preprocessing, models and thresholds avoids Variant-specific adaptation. Exact-profile overlap must also be checked: a separate CSV need not contain profiles unseen during Base development.

Random Forest randomisation is a modelling design (Breiman, 2001), not a theorem that bagged trees transfer better than boosted trees or transformers. A change in average precision may reflect both score ordering and the evaluated population, including prevalence. We therefore report retained cohorts, multiple metrics and the fixed operating rule.

### 2.3. SHAP and attention have different scopes

SHAP decomposes model output relative to an explainer reference (Lundberg and Lee, 2017; Lundberg, Erion, Chen, DeGrave, Prutkin, Nair, Katz, Himmelfarb, Bansal and Lee, 2020). It describes the fitted model, and depends on feature dependence, background choice and grouping (Molnar, 2025). Jaccard similarity of top- $k$  sets measures membership overlap:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

It does not compare ranking order, attribution magnitudes or causal effects.

Attention weights describe allocation within a network computation. A named feature token does not make its attention weight the feature’s effect on the final output. Head averaging may conceal structure; residual paths and feed-forward blocks are outside the inspected matrix. SHAP and attention are therefore separate analyses here, not independent measurements validating a shared causal construct.

### 3. Experimental Setup

#### 3.1. Datasets and evaluated populations

Raw ULB 2013 contains 284,807 transactions and 492 frauds (Machine Learning Group, Université Libre de Bruxelles, 2016; Dal Pozzolo, Caelen, Johnson and Bontempi, 2015). Exact predictor-profile deduplication before splitting retains 283,726 profiles and 473 frauds. A stratified seed-42 split assigns 226,980 profiles (378 frauds) to development (DEV) and 56,746 (95 frauds) to TEST. Predictors exclude the target; the 28 PCA components, transaction time and amount remain (30 predictors). Estimates concern distinct observed profiles, not the original transaction-frequency population. Customer independence is not inferred.

BAF Base contains 1,000,000 applications; DEV contains 800,000 and TEST 200,000, including 2,206 TEST frauds. We use 30 common predictors: 25 treated as numerical and five nominal categorical variables. The target and month are excluded; months are pooled. The extra  $x_1/x_2$  variables in Variants III/V are excluded to preserve the same schema.

From each original 200,000-row Variant TEST partition, every common-30 predictor profile matching a Base DEV profile is excluded. Numeric canonicalisation and element-wise confirmation supplement hash candidate matching; exclusions do not use Variant labels. Table 1 gives the populations shared by transfer and Variant SHAP. These

**Table 1**

BAF transfer cohorts excluding exact matches to Base DEV. Parentheses give excluded fraud cases.

| Cohort      | Original | Excluded     | Retained | Frauds | Prevalence (%) |
|-------------|----------|--------------|----------|--------|----------------|
| Base        | 200,000  | 0 (0)        | 200,000  | 2,206  | 1.1030         |
| Variant I   | 200,000  | 22,474 (262) | 177,526  | 1,944  | 1.0951         |
| Variant II  | 200,000  | 22,469 (250) | 177,531  | 1,956  | 1.1018         |
| Variant III | 200,000  | 21,282 (245) | 178,718  | 1,961  | 1.0973         |
| Variant IV  | 200,000  | 21,350 (246) | 178,650  | 1,960  | 1.0971         |
| Variant V   | 200,000  | 21,172 (228) | 178,828  | 1,978  | 1.1061         |

safeguards establish exact-profile separation from Base DEV, not entity independence, separation from Base TEST or mutual independence among Variants. Filtering changes the population; the common projection does not isolate full separability interventions involving  $x_1/x_2$ .

#### 3.2. Preprocessing, models and selection

The five supervised families are Logistic Regression (LR), Random Forest (RF), LightGBM (LGBM), CatBoost and a custom FT-Transformer. One-Class SVM (OCSVM) is a separate reference, fitted only on legitimate training observations; validation labels still select its threshold. It uses an RBF kernel,  $\nu = 0.01$  and  $\text{gamma}=\text{scale}$ , without Optuna tuning.

Numerical imputation and standardisation are fitted only within the relevant training partition. Classical pipelines one-hot encode nominal categories, including CatBoost; native CatBoost categorical processing is not evaluated. FT-Transformer uses learned numerical tokens and embeddings of integer category codes. This custom implementation is one representative of tabular transformers, not an assessment of every FT-Transformer implementation.

The primary BAF policy preserves documented numerical absence codes (Jesus et al., 2022a). Actual NaN values are mean-imputed, but coded absence enters standardisation as a numerical magnitude. A predefined sensitivity converts only documented absence codes to NaN, adds indicators and applies training-fitted mean imputation. It holds baseline hyperparameters fixed; the primary policy is not chosen retrospectively using TEST.

Classical supervised baselines use 50 attempted Optuna TPE trials (Akiba, Sano, Yanase, Ohta and Koyama, 2019), with pruning and five-fold stratified inner validation. Hyperparameters maximise validation AP and are reused across imbalance strategies. FT-Transformer uses one stratified 80/20 holdout within DEV instead of five-fold validation. ULB repeats searches on the distinct-profile partition; BAF retains its previously selected hyperparameters. Frozen classical BAF final models are preserved while validation evidence is recovered. BAF FT-Transformer follows the deterministic validation/final-training schedule with retained hyperparameters, without a new BAF search.

For the ULB search, FT-Transformer trial initialisation uses seed 42+trial number. A seed-42 validation refit selects checkpoint, threshold and epoch count using validation AP. A freshly initialised seed-42 model then fits all DEV for that epoch count. AdamW and cosine scheduling retain a 200-epoch horizon. The BAF baseline has 64-dimensional

tokens, two transformer blocks and eight heads. Validation scores, indices, preprocessors and checkpoints identify the executed path. The two validation designs protect TEST but do not estimate the same selection variability.

### 3.3. Imbalance and representation controls

The primary comparison uses None, random undersampling (RUS), random oversampling (ROS), SMOTE, SMOTE+Tomek, SMOTEENN and Class Weights. Samplers act only on training data, targeting class balance. Baseline hyperparameters are reused rather than retuned for every strategy.

Primary classical SMOTE interpolates standardised numerical and one-hot categorical columns. Primary FT-Transformer SMOTE interpolates numerical values and ordinal category codes, subsequently converting synthetic category values to integer embedding indices. It does not interpolate learned category embeddings. Neither path is categorical-aware. Separate fixed-hyperparameter SMOTE-NC controls for BAF CatBoost and FT-Transformer preserve the five nominal categories. Numerical binary variables remain in the interpolated numerical block, so not all discrete constraints are enforced. Controls are outside the seven-strategy matrix.

### 3.4. Metrics, thresholds and uncertainty

PR-AUC denotes average precision (AP), the primary ranking summary; ROC-AUC is complementary (Saito and Rehmsmeier, 2015). The operational rule maximises validation  $F_2$ :

$$F_2 = \frac{5TP}{5TP + 4FN + FP}.$$

For classical models, the final threshold is the median of five fold-selected thresholds. FT-Transformer uses saved internal validation scores. Final models fit all DEV; thresholds are frozen before TEST. No TEST fitting, probability calibration or threshold selection occurs. Threshold selection is not probability calibration.

Transfer uses saved Base models, fitted preprocessing and Base DEV-selected thresholds without Variant retraining or recalibration. Base is an in-domain reference, not an upper bound for Variant AP.

Baseline and transfer 95% intervals use 1,000 seed-42 TEST-row bootstrap resamples with fitted models and thresholds fixed; single-class draws are discarded. Primary strategy cells are point estimates. Representation controls use paired resampling with the same rows for both configurations. Intervals omit refitting, seed, split and entity-cluster variability. They are not simultaneous guarantees; overlapping marginal intervals or a paired interval containing zero does not establish equivalence. Close transfer values retain four decimals; primary strategy cells retain three. Summaries use complete saved scores and integer confusion counts before display rounding.

### 3.5. Compatible SHAP and separate attention

BAF SHAP sources are separately pinned to the same frozen LR, LGBM and CatBoost fitted functions used in predictive reporting. LinearExplainer explains LR; TreeExplainer explains the boosted models (Lundberg and Lee, 2017; Lundberg et al., 2020). LR uses an empirical preprocessed Base TEST reference, possibly summarised by its masker; tree explainers use their tree-path-dependent reference. This post-hoc use of unlabelled attributes neither fits the predictive model nor selects its threshold. Different references preclude direct cross-model comparison of absolute attribution magnitudes.

Global original-feature importance averages absolute contributions across all 200,000 Base TEST rows. For one-hot groups, absolute encoded contributions are summed before averaging; local explanations sum signed contributions to preserve additivity. FT-Transformer GradientExplainer is excluded: its wrapper converts floating-point categorical inputs to integer embedding indices, preventing reliable gradient attribution to those inputs.

Variant explanations use the frozen Base-trained LGBM on every retained row. Native tree-path-dependent TreeSHAP operates in raw log-odds and is checked against saved Base TreeExplainer contributions at 128 uniformly spaced TEST positions with absolute tolerance  $10^{-10}$ . Complete importance vectors precede top-10 selection.

Attention uses the baseline BAF FT-Transformer on all 200,000 Base TEST rows. It extracts the final-layer [CLS] query row, with PyTorch averaging across heads. Entropy includes all 31 tokens, including [CLS], and is normalised by  $\ln 31$ . Categorical/numerical attention compares mean weight per token, not total group mass. Diagnostic predictions reproduce saved scores and decisions. Case heatmaps select errors nearest the frozen threshold. Earlier layers and individual heads are outside the aggregate view.

## 4. Results

### 4.1. Imbalance sensitivity

Tables 2 and 3 compare FT-Transformer and CatBoost. Population standard deviation across seven strategies describes intervention spread, not seed variation or significance.

**Table 2**

FT-Transformer and CatBoost on distinct ULB profiles. Final row: population standard deviation across seven complete strategies, calculated before display rounding.

| Strategy              | FT-Transformer |              | CatBoost     |              |
|-----------------------|----------------|--------------|--------------|--------------|
|                       | PR-AUC         | $F_2$        | PR-AUC       | $F_2$        |
| None                  | 0.801          | 0.761        | 0.820        | 0.775        |
| RUS                   | 0.641          | 0.600        | 0.637        | 0.763        |
| ROS                   | 0.784          | 0.763        | 0.811        | 0.808        |
| SMOTE                 | 0.785          | 0.692        | 0.798        | 0.788        |
| SM+Tomek              | 0.785          | 0.692        | 0.798        | 0.788        |
| SMOTEENN              | 0.797          | 0.698        | 0.793        | 0.779        |
| Weights               | 0.756          | 0.774        | 0.802        | 0.776        |
| <i>Pop. std. dev.</i> | <i>0.052</i>   | <i>0.057</i> | <i>0.059</i> | <i>0.013</i> |

**Table 3**

FT-Transformer and CatBoost on BAF Base; same standard-deviation definition as Table 2.

| Strategy              | FT-Transformer |              | CatBoost     |              |
|-----------------------|----------------|--------------|--------------|--------------|
|                       | PR-AUC         | $F_2$        | PR-AUC       | $F_2$        |
| None                  | 0.177          | 0.304        | 0.180        | 0.314        |
| RUS                   | 0.163          | 0.301        | 0.178        | 0.317        |
| ROS                   | 0.166          | 0.310        | 0.185        | 0.316        |
| SMOTE                 | 0.115          | 0.257        | 0.154        | 0.303        |
| SM+Tomek              | 0.115          | 0.257        | 0.154        | 0.303        |
| SMOTEENN              | 0.124          | 0.269        | 0.165        | 0.309        |
| Weights               | 0.176          | 0.313        | 0.184        | 0.317        |
| <i>Pop. std. dev.</i> | <i>0.027</i>   | <i>0.023</i> | <i>0.012</i> | <i>0.006</i> |

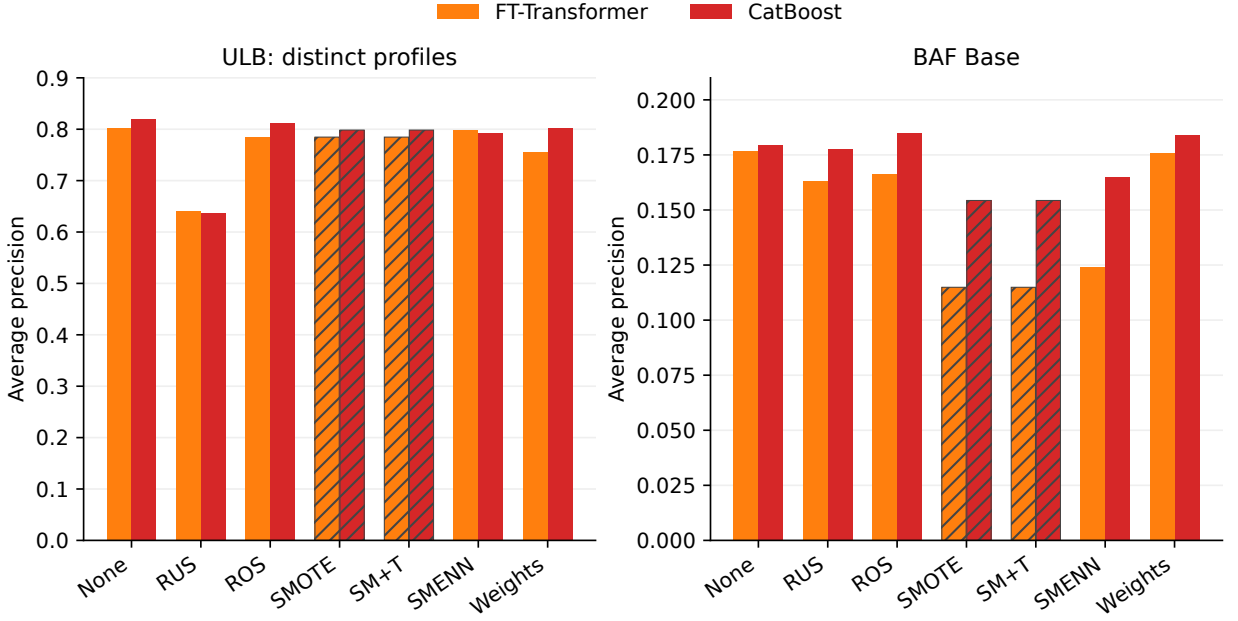
On ULB, RUS produces the largest AP reduction for both models. CatBoost has higher AP except under RUS and SMOTEENN. AP spread is 0.059 for CatBoost versus 0.052 for FT-Transformer, but  $F_2$  spread reverses: 0.013 versus 0.057. Lower dispersion under one metric is not a general robustness advantage. Class Weights reduces transformer AP on ULB while increasing  $F_2$  slightly; the metric and decision rule matter.

BAF baseline AP is 0.1769 for FT-Transformer and 0.1796 for CatBoost, without established equivalence. SMOTE reduces these to 0.1149 and 0.1543: approximately 35% and 14% below baseline. Class Weights is less than 1% below transformer baseline AP and increases  $F_2$  from 0.304 to 0.313, but is not universally safest across datasets. CatBoost ROS reaches 0.185 AP. Figure 1 displays the primary profiles.

SMOTE and SMOTE+Tomek are identical effective treatments in logged revised fitting stages: Tomek removes zero rows, with identical inputs and TEST scores. Historical BAF classical scores also coincide up to floating-point noise. A reconstructed full-DEV sampler removes zero rows but does not replace unavailable historical fold logs. This explains equality here, not algorithmic equivalence generally.

### 4.2. Predefined representation controls

Table 4 separates primary SMOTE from the categorical-aware control. Transformer SMOTE-NC reaches AP 0.1157, well below the 0.1769 baseline. The paired change is +0.0008 with interval  $[-0.0068, +0.0076]$ ; it does not establish equivalence to primary SMOTE. Its alert rate changes from 6.11% to 4.22% under separately DEV-selected thresholds, despite similar AP and  $F_2$ . CatBoost has lower AP and  $F_2$  in the tested control.



**Figure 1:** Primary AP point estimates across seven strategies. Hatched bars mark SMOTE and SMOTE+Tomek. Differences describe implemented sampling representations, not an isolated architectural mechanism.

**Table 4**

Categorical-aware BAF controls. Differences are SMOTE-NC minus primary SMOTE, with paired 95% row-bootstrap intervals ( $B = 1,000$ ).

| Model          | Metric | Primary | Control | $\Delta$ [paired 95% CI]   |
|----------------|--------|---------|---------|----------------------------|
| CatBoost       | AP     | 0.1543  | 0.1433  | -0.0110 [-0.0147, -0.0074] |
|                | $F_2$  | 0.3029  | 0.2856  | -0.0173 [-0.0233, -0.0108] |
| FT-Transformer | AP     | 0.1149  | 0.1157  | +0.0008 [-0.0068, +0.0076] |
|                | $F_2$  | 0.2567  | 0.2577  | +0.0010 [-0.0088, +0.0111] |

The intervention changes representation and nominal-aware synthesis jointly; it isolates neither architecture nor an optimally tuned alternative. Historical primary CatBoost SMOTE retains its saved six-decimal threshold and reproduced confusion counts. Pairing verifies raw-file identity, seed-42 splitting and identical ordered TEST labels, without assuming an unavailable historical TEST-index artefact.

The absence-code sensitivity has mixed AP directions (Table 5): LR gains 0.0045, while OCSVM loses 0.0028. Remaining AP difference intervals include zero, as do all supervised  $F_2$  difference intervals. These findings neither select an optimal policy nor establish equal policies.

### 4.3. Controlled Base-to-Variant transfer

Tables 6–8 report full Base TEST and the same filtered Variants for every model. Bold marks each model’s highest cohort value, not the winner within a Variant.

LGBM leads AP on Variant I (0.1483); CatBoost leads Variants II–V (0.1822, 0.1329, 0.1668 and 0.1207). The  $F_2$  leaders differ: CatBoost leads Variant I (0.2701), while LGBM leads II–V (0.3035, 0.2760, 0.3008 and 0.2620). These are point-estimate rankings, not universal or formally established pairwise superiority. RF AP is below Base on four Variants and slightly higher on II (0.1615 versus 0.1588), with no general transfer advantage. All supervised  $F_2$  values fall on every Variant at frozen Base thresholds.

Variant II AP is slightly above Base for the four classical supervised models and slightly below for FT-Transformer. An increase is compatible with verified profile separation: ranking and population can change in either direction.

**Table 5**

Absence-code sensitivity. Differences are NaN conversion, mean imputation and indicators minus preserved codes, with paired 95% row-bootstrap intervals ( $B = 1,000$ ).

| Model          | Metric | Primary | Control | $\Delta$ [paired 95% CI]   |
|----------------|--------|---------|---------|----------------------------|
| LR             | AP     | 0.1432  | 0.1477  | +0.0045 [+0.0017, +0.0074] |
|                | $F_2$  | 0.2868  | 0.2910  | +0.0042 [−0.0008, +0.0096] |
| RF             | AP     | 0.1588  | 0.1596  | +0.0008 [−0.0015, +0.0030] |
|                | $F_2$  | 0.2941  | 0.2931  | −0.0010 [−0.0061, +0.0044] |
| LGBM           | AP     | 0.1768  | 0.1759  | −0.0009 [−0.0029, +0.0011] |
|                | $F_2$  | 0.3159  | 0.3173  | +0.0015 [−0.0039, +0.0062] |
| CatBoost       | AP     | 0.1796  | 0.1790  | −0.0005 [−0.0023, +0.0015] |
|                | $F_2$  | 0.3143  | 0.3126  | −0.0017 [−0.0063, +0.0031] |
| FT-Transformer | AP     | 0.1769  | 0.1732  | −0.0038 [−0.0082, +0.0006] |
|                | $F_2$  | 0.3041  | 0.3026  | −0.0016 [−0.0068, +0.0034] |
| OCSVM          | AP     | 0.0190  | 0.0162  | −0.0028 [−0.0035, −0.0024] |
|                | $F_2$  | 0.0805  | 0.0679  | −0.0126 [−0.0166, −0.0087] |

**Table 6**

Transferred PR-AUC (average precision), without retraining.

| Model     | Base          | Var I  | Var II        | Var III | Var IV | Var V  |
|-----------|---------------|--------|---------------|---------|--------|--------|
| LR        | 0.1432        | 0.1066 | <b>0.1442</b> | 0.1039  | 0.1334 | 0.1005 |
| RF        | 0.1588        | 0.1342 | <b>0.1615</b> | 0.1193  | 0.1477 | 0.1093 |
| LGBM      | 0.1768        | 0.1483 | <b>0.1803</b> | 0.1328  | 0.1646 | 0.1185 |
| CatBoost  | 0.1796        | 0.1472 | <b>0.1822</b> | 0.1329  | 0.1668 | 0.1207 |
| FT-Trans. | <b>0.1769</b> | 0.1462 | 0.1751        | 0.1327  | 0.1589 | 0.1168 |
| OCSVM     | 0.0190        | 0.0164 | <b>0.0220</b> | 0.0178  | 0.0205 | 0.0165 |

**Table 7**

Transferred  $F_2$  at frozen Base DEV-selected thresholds.

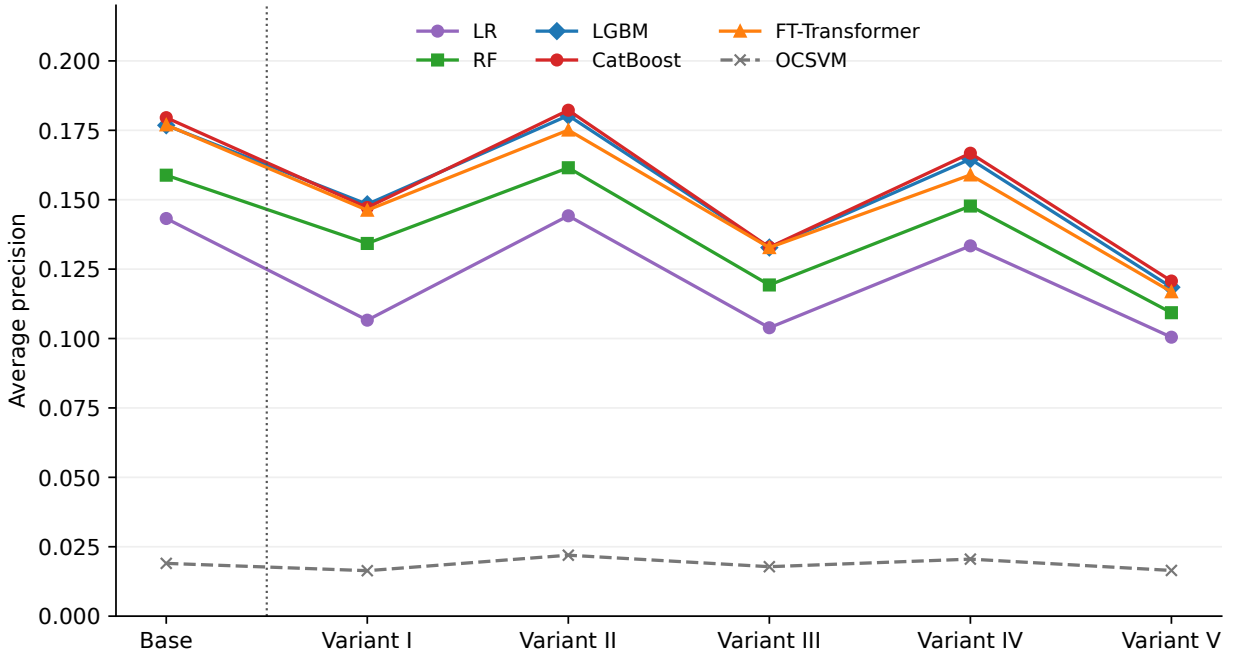
| Model     | Base          | Var I  | Var II        | Var III | Var IV | Var V  |
|-----------|---------------|--------|---------------|---------|--------|--------|
| LR        | <b>0.2868</b> | 0.2391 | 0.2656        | 0.2370  | 0.2680 | 0.2284 |
| RF        | <b>0.2941</b> | 0.2459 | 0.2840        | 0.2522  | 0.2796 | 0.2438 |
| LGBM      | <b>0.3159</b> | 0.2693 | 0.3035        | 0.2760  | 0.3008 | 0.2620 |
| CatBoost  | <b>0.3143</b> | 0.2701 | 0.2994        | 0.2697  | 0.2991 | 0.2588 |
| FT-Trans. | <b>0.3041</b> | 0.2661 | 0.2655        | 0.2489  | 0.2722 | 0.2391 |
| OCSVM     | 0.0805        | 0.0662 | <b>0.0925</b> | 0.0749  | 0.0870 | 0.0690 |

**Table 8**

Transferred ROC-AUC under the common 30-feature projection.

| Model     | Base   | Var I  | Var II        | Var III | Var IV        | Var V  |
|-----------|--------|--------|---------------|---------|---------------|--------|
| LR        | 0.8745 | 0.8488 | 0.8786        | 0.8517  | <b>0.8819</b> | 0.8450 |
| RF        | 0.8778 | 0.8533 | 0.8851        | 0.8586  | <b>0.8880</b> | 0.8461 |
| LGBM      | 0.8985 | 0.8820 | 0.9004        | 0.8809  | <b>0.9062</b> | 0.8710 |
| CatBoost  | 0.8980 | 0.8812 | 0.8994        | 0.8797  | <b>0.9042</b> | 0.8712 |
| FT-Trans. | 0.8965 | 0.8785 | 0.8964        | 0.8760  | <b>0.9009</b> | 0.8681 |
| OCSVM     | 0.6279 | 0.5959 | <b>0.6616</b> | 0.6144  | 0.6456        | 0.5907 |

Fraud models beyond in-domain performance



**Figure 2:** Base and retained-Variant AP. The dashed separator distinguishes in-domain evaluation from controlled transfer. LGBM leads Variant I and CatBoost leads II–V by point estimate.

Variants III/V produce approximately 25–34% reductions for the leading Base models, but omitted separability variables, pooled months and filtering prevent an isolated causal explanation. Supervised Variant ROC-AUC spans approximately 0.845–0.906. ROC-AUC is insensitive to pure class-prior changes with fixed class-conditional score distributions, but those distributions may change here. OCSVM stays below 0.023 AP on each Variant; this bounds its performance here, not anomaly detection generally.

#### 4.4. Compatible SHAP feature sets

Figure 3 shows each compatible model’s own top 15 original features. LR, LGBM and CatBoost share the leading trio `device_os`, `housing_status` and `phone_home_valid`. Absolute magnitudes are not directly comparable across explainer references.

LGBM and CatBoost share all top-ten members (Jaccard 1.00). Each shares eight with LR, giving  $8/(10+10-8) = 0.67$  (Table 9).

**Table 9**

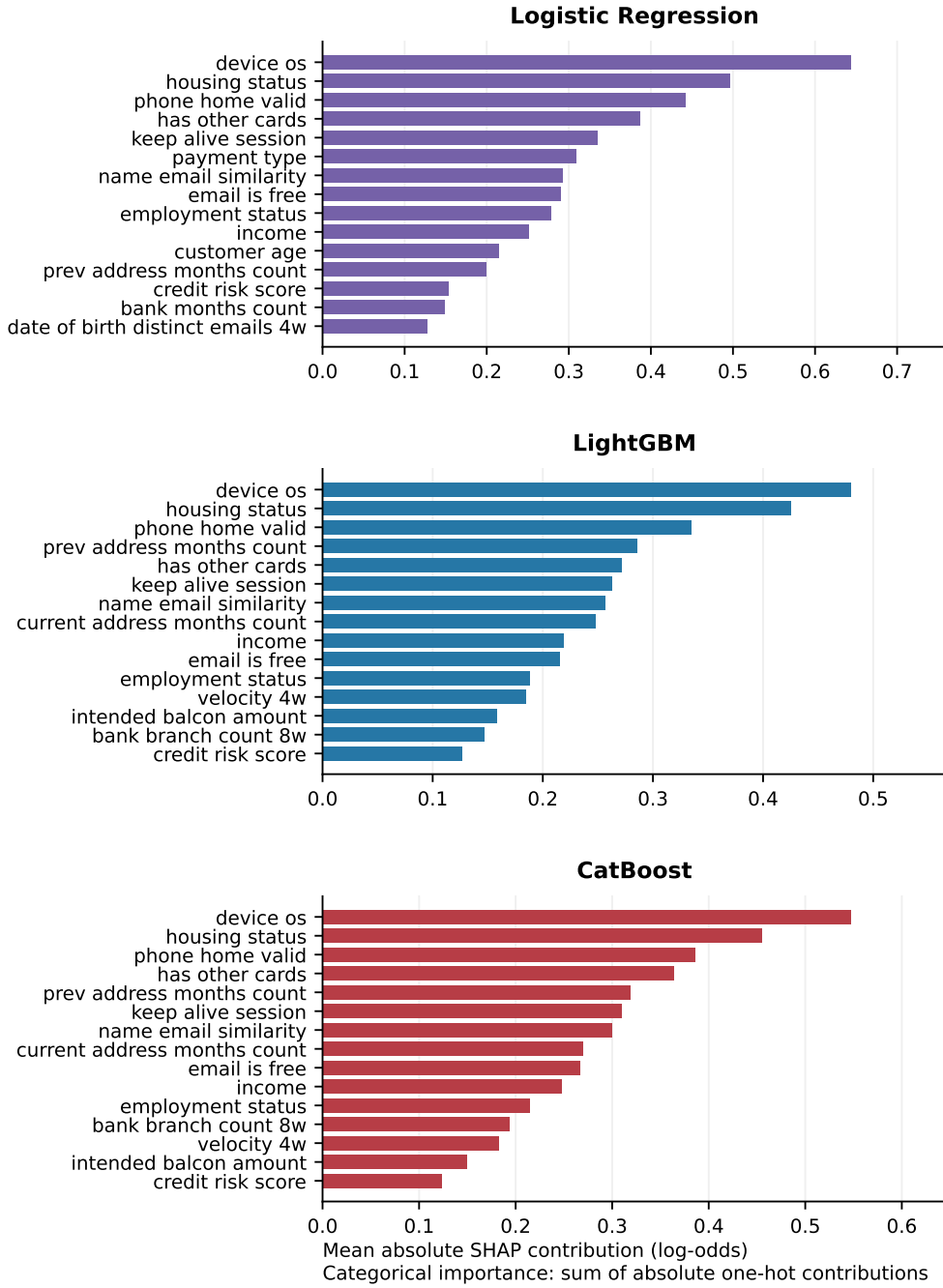
Cross-model Jaccard agreement of top-10 original-feature SHAP sets.

|          | LR   | LGBM | CatBoost |
|----------|------|------|----------|
| LR       | —    | 0.67 | 0.67     |
| LGBM     | 0.67 | —    | 1.00     |
| CatBoost | 0.67 | 1.00 | —        |

The fixed Base-trained LGBM retains the same top-10 set across all cohorts (Table 10). Rankings are not identical: `has_other_cards` and `prev_address_months_count` exchange fourth/fifth positions between Base/Variant I and II–V. Magnitudes also vary. Stable membership at one  $k$  is not an invariant feature–target relationship, unchanged local explanations or guaranteed validity under shift.

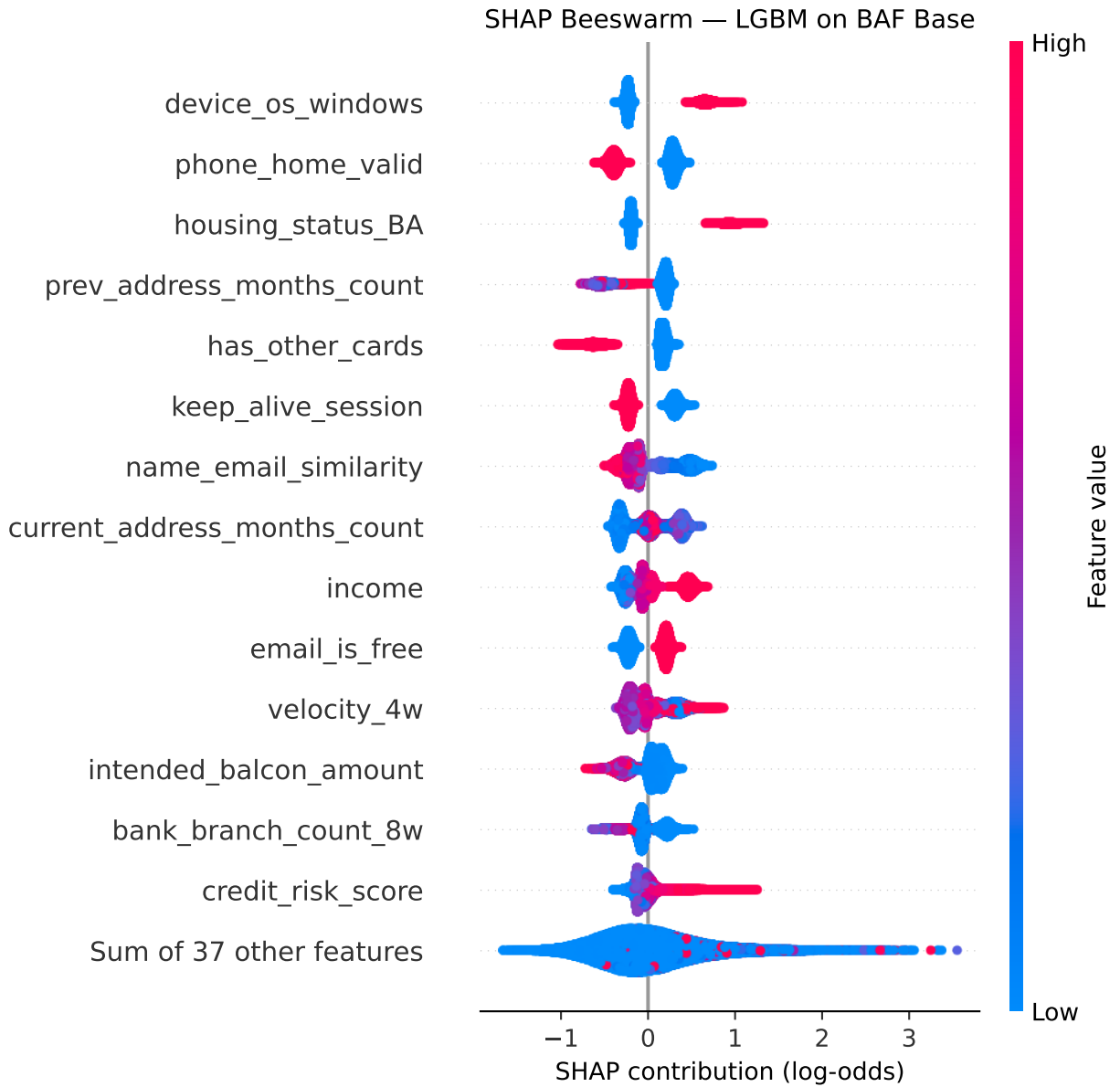
The complete signed decomposition preserves additivity; the residual contains all omitted contributions. These deliberately selected extremes illustrate model behaviour, not representative error-class estimates. A false-positive

### Fraud models beyond in-domain performance



**Figure 3:** Each compatible model's own top-15 original-feature importance, in log-odds. Categorical importance sums absolute one-hot contributions before averaging. FT-Transformer gradient SHAP is excluded.

explanation describes a high score without validating it. The lowest-score FN is not borderline, and these examples do not quantify the workload needed to recover all missed frauds.



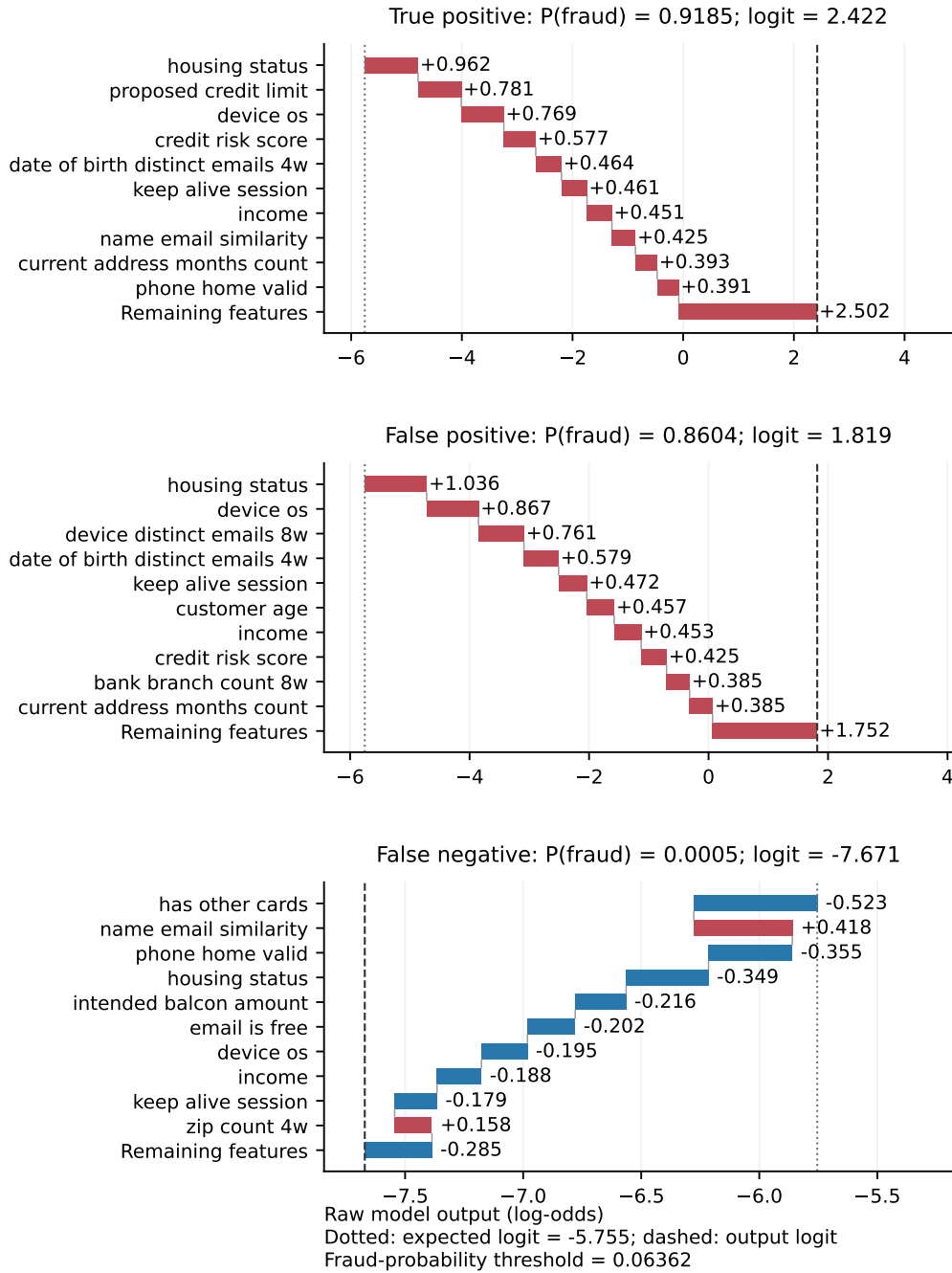
**Figure 4:** LGBM SHAP beeswarm in log-odds: 14 leading encoded columns and the signed sum of the remaining 37. Colour denotes transformed feature values, not causal fraud effects.

**Table 10**

Jaccard agreement for frozen LGBM top-10 original-feature sets on Base TEST and filtered Variant cohorts.

|          | Base | Var. I | Var. II | Var. III | Var. IV | Var. V |
|----------|------|--------|---------|----------|---------|--------|
| Base     | —    | 1.00   | 1.00    | 1.00     | 1.00    | 1.00   |
| Var. I   | 1.00 | —      | 1.00    | 1.00     | 1.00    | 1.00   |
| Var. II  | 1.00 | 1.00   | —       | 1.00     | 1.00    | 1.00   |
| Var. III | 1.00 | 1.00   | 1.00    | —        | 1.00    | 1.00   |
| Var. IV  | 1.00 | 1.00   | 1.00    | 1.00     | —       | 1.00   |
| Var. V   | 1.00 | 1.00   | 1.00    | 1.00     | 1.00    | —      |

## Fraud models beyond in-domain performance

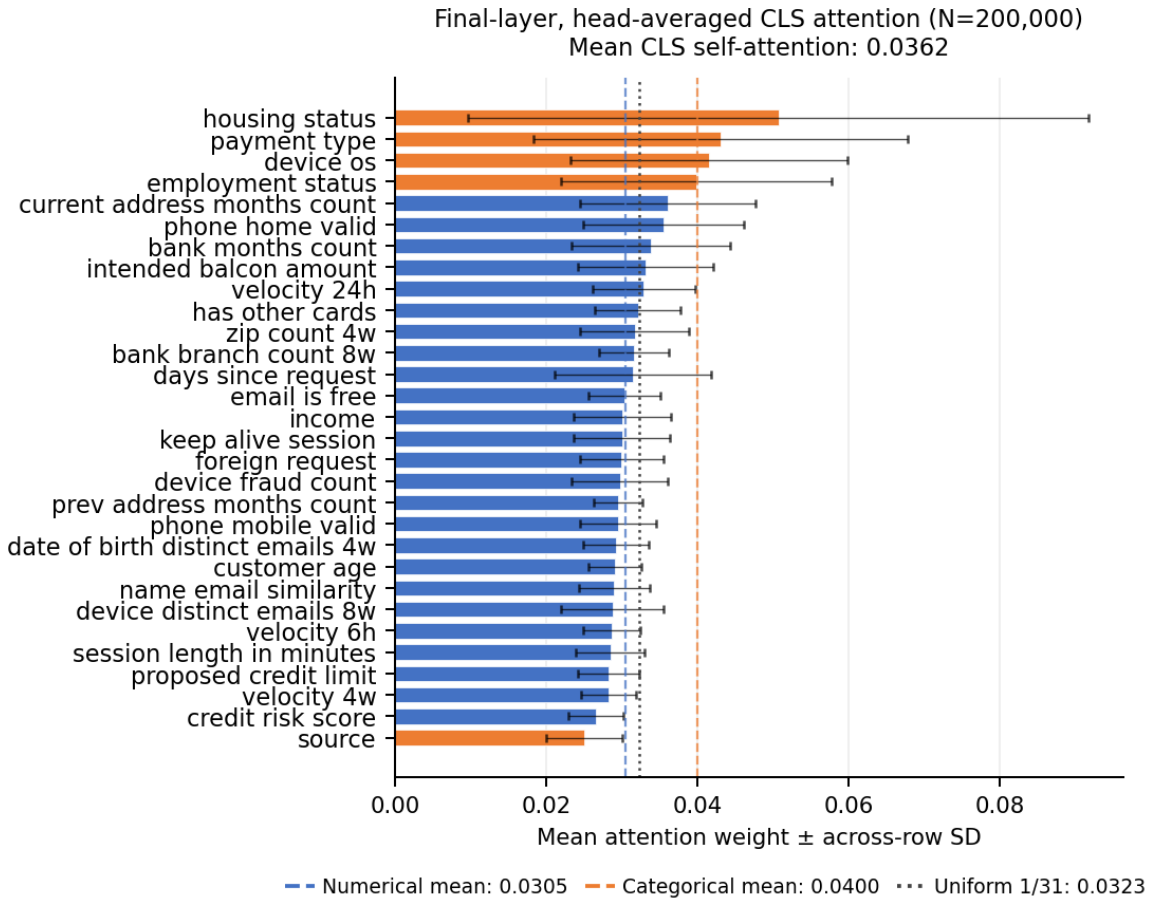


**Figure 5:** Additive LGBM log-odds waterfalls for highest-score TP and FP and lowest-score FN. Ten signed original-feature contributions plus the residual connect expected and case outputs; red increases the output towards fraud.

### 4.5. Separate FT-Transformer attention diagnostics

Mean normalised entropy is 0.985 (mean entropy 3.381 nats;  $\ln 31 = 3.434$ ). Mean [CLS] self-attention is 3.6%, compared with a uniform reference of about 3.2%. Mean weight per categorical/numerical token is 4.0%/3.1%; the unrounded ratio is 1.311. These descriptive statistics do not define formal pathology thresholds.

Largest mean feature-token weights are `housing_status` (5.1%), `payment_type` (4.3%) and `device_os` (4.2%). Aggregate [CLS] attention is diffuse; individual queries, heads and earlier layers need not be. Housing status and



**Figure 6:** Final-layer, head-averaged [CLS] attention on 200,000 BAF Base TEST rows. Error bars are standard deviations across samples; uniform reference: 1/31. Group means are not total group mass.

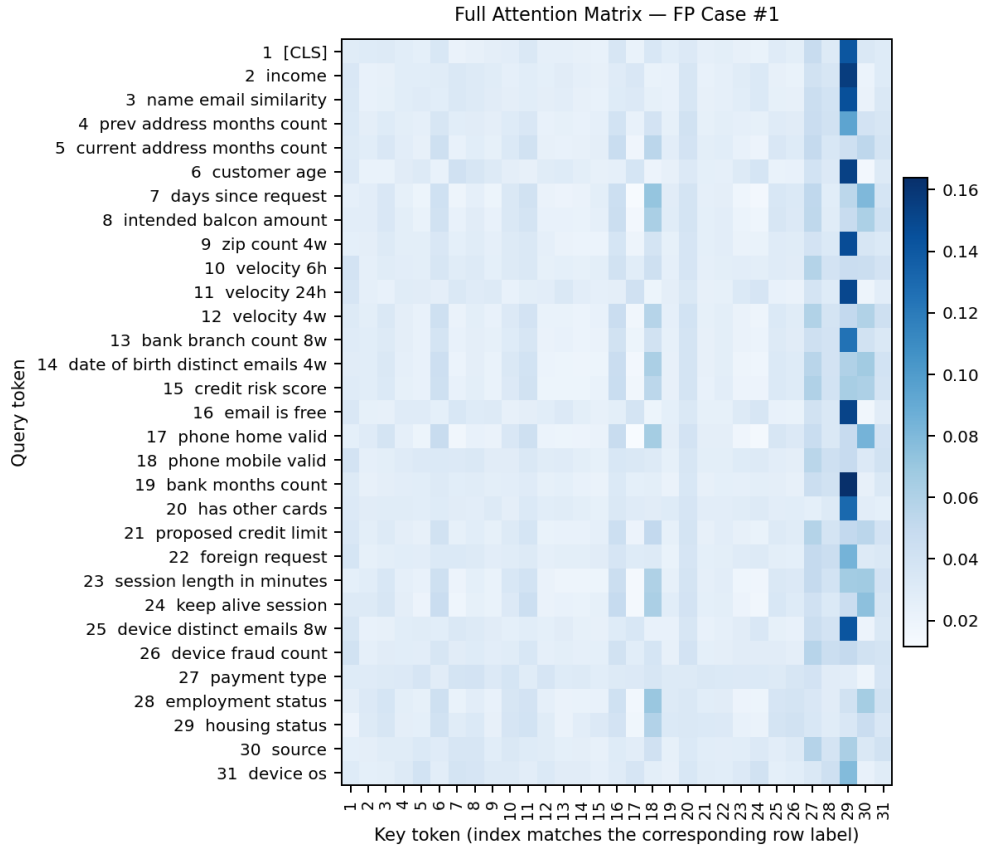
**Table 11**

Recorded BAF baseline costs (seconds). †: historical measurements retained with frozen classical models. Revised FT-Transformer has no new search; validation recovery is separate.

| Model     | Tuning (s)           | Train (s)           | Inference (s)       |
|-----------|----------------------|---------------------|---------------------|
| LR        | 55,070 <sup>†</sup>  | 83.9 <sup>†</sup>   | 0.209               |
| RF        | 105,313 <sup>†</sup> | 1494.0 <sup>†</sup> | 10.642              |
| LGBM      | 1,243 <sup>†</sup>   | 16.1 <sup>†</sup>   | 1.298               |
| CatBoost  | 7,010 <sup>†</sup>   | 36.7 <sup>†</sup>   | 0.213               |
| FT-Trans. | —                    | 354.4               | 4.262               |
| OCSVM     | —                    | 919.6 <sup>†</sup>  | 71.579 <sup>†</sup> |

device operating system also appear among leading SHAP features, but overlap establishes neither faithful attention attribution nor cross-method causal validity.

The matrix has an elevated housing-status key column for several queries. This selected error does not show uniform diffusion for every query or error. Diagnostic scores have zero threshold disagreements over 200,000 TEST rows, verifying extraction fidelity, not explanatory faithfulness. The weights neither explain close baseline AP values nor bound possible improvements from another model, representation or feature set.



**Figure 7:** Final-layer, head-averaged  $31 \times 31$  matrix for the FP nearest the frozen threshold ( $p \approx \tau = 0.0584$ ). Cells are query-to-key weights; case-specific elevations need not resemble aggregate [CLS] attention.

#### 4.6. Recorded computational burden

LGBM has lower recorded final-fit/inference times than FT-Transformer: 16.1/1.298 versus 354.4/4.262 seconds. Classical models execute on CPU and the transformer on GPU on an i5-13600KF/16 GB/RTX 3070 machine. Historical/revised executions span Python 3.12.7/3.12.14 and model-specific validation work; concurrent workloads and realised fit lengths also matter. There is no new BAF transformer search time comparable with retained classical tuning times. Records describe resource burden, not an intrinsic speed ratio or matched-hardware benchmark.

### 5. Discussion

#### 5.1. Model and representation are jointly evaluated

The larger transformer SMOTE reduction on BAF is a pipeline difference. Both primary paths alter encoded features using different category representations. Integer conversion of interpolated ordinal values is not interpolation of learned embeddings; trees also respond to altered training representations. The categorical-aware control does not recover baseline transformer AP, but changes representation and synthesis jointly, retains fixed hyperparameters and leaves numerical binary constraints unenforced. It cannot identify a unique architectural failure.

Absence-code sensitivity jointly changes conversion, imputation and indicators. Its mixed effects justify explicit representation reporting, not a universal policy. The primary policy stays predefined. Class Weights results likewise do not establish an untested objective, such as focal loss, as an improvement. New objectives and independently tuned categorical-aware pipelines remain future-work candidates.

## 5.2. Transfer depends on metric and population

LGBM and CatBoost lead transferred point estimates, with the leader depending on metric and Variant. That is not a generic advantage under every shift. The synthetic populations, pooled months, omitted separability features and profile filtering define the scope.

Every supervised  $F_2$  falls at frozen thresholds even where AP rises. Preserved ranking quality therefore does not guarantee preserved operational performance. Freezing Base thresholds is the experiment, not proof that they are optimal or stable on every Variant. Prospective use would require appropriate temporal or external evaluation and a separately validated decision rule.

## 5.3. Interpretation agreement is bounded

Compatible global SHAP sets provide common candidate features for inspection. Stable top-10 membership for fixed LGBM is useful descriptive evidence, not a waiver of monitoring, local explanation review or operational revalidation. Neither common membership nor additivity establishes causality.

The attention diagnostic asks how concentrated one final-layer head-averaged allocation is. Diffusion is not evidence that the inputs contain no additional signal. It cannot identify an information-theoretic AP ceiling, diagnose all pathological behaviour or rule out improvements from other architectures. Comparing model reliance would require compatible output-sensitive methods and validation of their relationship to predictions.

## 5.4. Limitations and reproducibility

The study uses one outer split and saved fitted models, not repeated seeds. Classical five-fold validation and transformer holdout estimate different selection variability. Equal attempted trial budgets do not imply equal compute or search coverage; strategies are not independently retuned.

ULB distinct-profile estimates do not concern its original transaction-frequency population; PCA features limit domain interpretation. BAF's shared synthetic process and possible TEST overlap limit independent deployment inference. Exact-profile safeguards do not prove entity independence. Row-bootstrap intervals omit training and cluster variability; multiple comparisons are not adjusted as simultaneous guarantees.

Interpretation depends on models, explainer references, grouping, top- $k$  choice and attention aggregation. The custom transformer is the sole deep-learning representative. Mixed execution provenance prevents intrinsic cost comparison.

A dated manifest pins configurations, scores, indices, thresholds, transfer cohorts, compatible explanations and reporting assets. Article tables are extracted from verified outputs; plots read saved evidence without fitting. Traceability complements, rather than substitutes for, independent empirical replication.

## 6. Conclusion

Tree ensembles remain strong candidates in this setting, but the evidence supports conditional comparison rather than universal sufficiency. Primary SMOTE has a larger BAF AP reduction for FT-Transformer than CatBoost; a fixed categorical-aware control does not restore its baseline. Representation remains a confound.

After exact-profile filtering, LGBM leads AP on Variant I and CatBoost on II–V. Their  $F_2$  leadership differs by Variant, and all supervised models lose  $F_2$  at frozen thresholds. Compatible SHAP comparisons establish stable global membership, not identical rankings, causality or deployment-ready explanations. Diffuse final-layer attention is a separate diagnostic, not exhausted signal.

The practical contribution is a traceable model–representation–threshold comparison on controlled populations. Future work should examine repeated seeds, temporal/external validation, independently tuned categorical-aware pipelines, earlier layers and individual heads, and compatible output-sensitive explanations. These extensions test unresolved questions without presuming an architectural mechanism or data-imposed ceiling.

## CRedit authorship contribution statement

**Olavo Caixeiro:** Conceptualization, Methodology, Software, Formal analysis, Writing – original draft.

**Maryam Abbasi:** Conceptualization, Supervision, Writing – review & editing.

**Pedro Miguel de Oliveira Martins:** Supervision, Writing – review & editing.

## Data Availability

ULB is available from Machine Learning Group, Université Libre de Bruxelles (2016) and BAF from Feedzai Research (2022), subject to their respective terms. Project code, selected aggregate evidence and the validated MSc dissertation are available at <https://github.com/0lavo200100274/FraudDetection>. Corrected evidence is identified by the dated 20261005 manifest. Fitted weights and row-level score, attribution and attention arrays are not distributed in the public scientific export. The public aggregate evidence does not replace those internal artefacts for exact saved-run reconstruction. The article PDF remains an unpublished manuscript. No publication or acceptance status is asserted for the companion manuscript.

## Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

OpenAI Codex was used to assist with manuscript revision, checks against the saved experimental evidence, and reproducible plotting and layout code for the figures. Experimental values were read from recorded evaluations, not invented or inferred by a generative model. The authors remain responsible for the accuracy, interpretation and final approval of the manuscript.

## References

- Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M., 2019. Optuna: A next-generation hyperparameter optimization framework, in: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, pp. 2623–2631. URL: <https://doi.org/10.1145/3292500.3330701>, doi:10.1145/3292500.3330701.
- Batista, G.E.A.P.A., Prati, R.C., Monard, M.C., 2004. A study of the behavior of several methods for balancing machine learning training data. ACM SIGKDD Explorations Newsletter 6, 20–29. URL: <https://doi.org/10.1145/1007730.1007735>, doi:10.1145/1007730.1007735.
- Borisov, V., Leemann, T., Seßler, K., Haug, J., Pawelczyk, M., Kasneci, G., 2024. Deep neural networks and tabular data: A survey. IEEE Transactions on Neural Networks and Learning Systems 35, 7499–7519. URL: <https://doi.org/10.1109/TNNLS.2022.3229161>, doi:10.1109/TNNLS.2022.3229161.
- Breiman, L., 2001. Random forests. Machine Learning 45, 5–32. URL: <https://doi.org/10.1023/A:1010933404324>, doi:10.1023/A:1010933404324.
- Caixeiro, O., Abbasi, M., Martins, P.M.d.O., 2026. Threshold selection as a critical design choice in financial fraud detection: A systematic comparison of classical machine learning models and tabular transformers. Unpublished companion manuscript.
- Chawla, N.V., Bowyer, K.W., Hall, L.O., Kegelmeyer, W.P., 2002. SMOTE: Synthetic minority over-sampling technique. Journal of Artificial Intelligence Research 16, 321–357. URL: <https://doi.org/10.1613/jair.953>, doi:10.1613/jair.953.
- Dal Pozzolo, A., Caelen, O., Johnson, R.A., Bontempi, G., 2015. Calibrating probability with undersampling for unbalanced classification, in: 2015 IEEE Symposium Series on Computational Intelligence (SSCI), pp. 159–166. URL: <https://doi.org/10.1109/SSCI.2015.33>, doi:10.1109/SSCI.2015.33.
- Feedzai Research, 2022. Bank account fraud dataset suite. URL: <https://github.com/feedzai/bank-account-fraud>. dataset repository; accessed 22 July 2026.
- Gorishniy, Y., Rubachev, I., Khrulkov, V., Babenko, A., 2021. Revisiting deep learning models for tabular data, in: Advances in Neural Information Processing Systems, pp. 18932–18943. URL: <https://papers.nips.cc/paper/2021/hash/9d86d83f925f2149e9edb0ac3b49229c-A-abstract.html>.
- Grinsztajn, L., Oyallon, E., Varoquaux, G., 2022. Why do tree-based models still outperform deep learning on typical tabular data?, in: Advances in Neural Information Processing Systems. URL: <https://doi.org/10.52202/068431-0037>, doi:10.52202/068431-0037. datasets and Benchmarks Track.
- Hilal, W., Gadsden, S.A., Yawney, J., 2022. Financial fraud: A review of anomaly detection techniques and recent advances. Expert Systems With Applications 193, 116429. URL: <https://doi.org/10.1016/j.eswa.2021.116429>, doi:10.1016/j.eswa.2021.116429.
- Jesus, S., Pombal, J., Alves, D., Cruz, A.F., Saleiro, P., Ribeiro, R.P., Gama, J., Bizarro, P., 2022a. BAF dataset suite datasheet. URL: <https://github.com/feedzai/bank-account-fraud/blob/main/documents/datasheet.pdf>. official dataset documentation accompanying the NeurIPS 2022 Datasets and Benchmarks paper; 8 pages; accessed 5 October 2026.
- Jesus, S., Pombal, J., Alves, D., Cruz, A.F., Saleiro, P., Ribeiro, R.P., Gama, J., Bizarro, P., 2022b. Turning the tables: Biased, imbalanced, dynamic tabular datasets for ml evaluation, in: Advances in Neural Information Processing Systems, pp. 33563–33575. URL: <https://doi.org/10.52202/068431-2432>, doi:10.52202/068431-2432. datasets and Benchmarks Track.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.Y., 2017. LightGBM: A highly efficient gradient boosting decision tree, in: Advances in Neural Information Processing Systems. URL: [https://proceedings.neurips.cc/paper\\_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html](https://proceedings.neurips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html).
- Lundberg, S.M., Erion, G., Chen, H., DeGrave, A., Prutkin, J.M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., Lee, S.I., 2020. From local explanations to global understanding with explainable AI for trees. Nature Machine Intelligence 2, 56–67. URL: <https://doi.org/10.1038/s42256-019-0138-9>, doi:10.1038/s42256-019-0138-9.
- Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems, pp. 4765–4774. URL: <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>.

- Machine Learning Group, Université Libre de Bruxelles, 2016. Credit card fraud detection. URL: <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>. dataset; observations collected in September 2013; first released on Kaggle in 2016; accessed 22 July 2026.
- Molnar, C., 2025. Interpretable Machine Learning: A Guide for Making Black Box Models Explainable. 3 ed., Christoph Molnar. URL: <https://christophm.github.io/interpretable-ml-book/>.
- Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V., Gulin, A., 2018. CatBoost: unbiased boosting with categorical features, in: Advances in Neural Information Processing Systems. URL: <https://proceedings.neurips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html>.
- Ryman-Tubb, N.F., Krause, P., Garn, W., 2018. How artificial intelligence and machine learning research impacts payment card fraud detection: A survey and industry benchmark. Engineering Applications of Artificial Intelligence 76, 130–157. URL: <https://doi.org/10.1016/j.engappai.2018.07.008>, doi:10.1016/j.engappai.2018.07.008.
- Saito, T., Rehmsmeier, M., 2015. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLOS ONE 10, e0118432. URL: <https://doi.org/10.1371/journal.pone.0118432>, doi:10.1371/journal.pone.0118432.
- Shwartz-Ziv, R., Armon, A., 2022. Tabular data: Deep learning is not all you need. Information Fusion 81, 84–90. URL: <https://doi.org/10.1016/j.inffus.2021.11.011>, doi:10.1016/j.inffus.2021.11.011.